Operating System Capstone

Lab 3: Memory Allocator

Sean, Hsien

Overview

You have 2 exercises on this lab

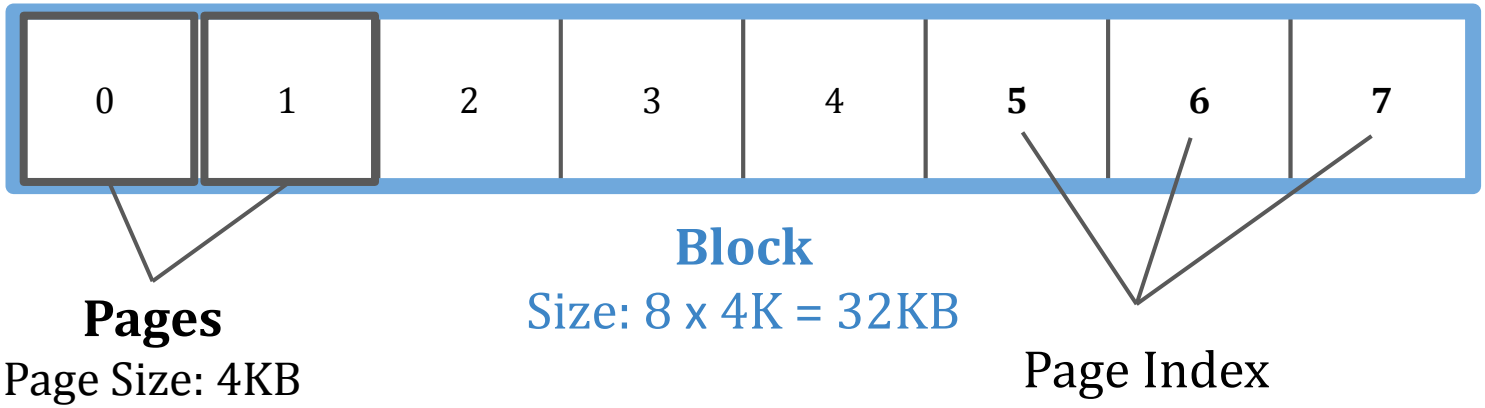
- Exercise 3-1: The Buddy Allocator
- Exercise 3-2: Memory Reservation

Exercise 3-1:Buddy System

1. **Search mechanism :** Locate the lowest available block in the free_area array starting from target level k >= order
2. **Split logic :** Bisect blocks iteratively when k > order
3. Downgrade and insert unallocated split remnants into the corresponding **free_area** lists

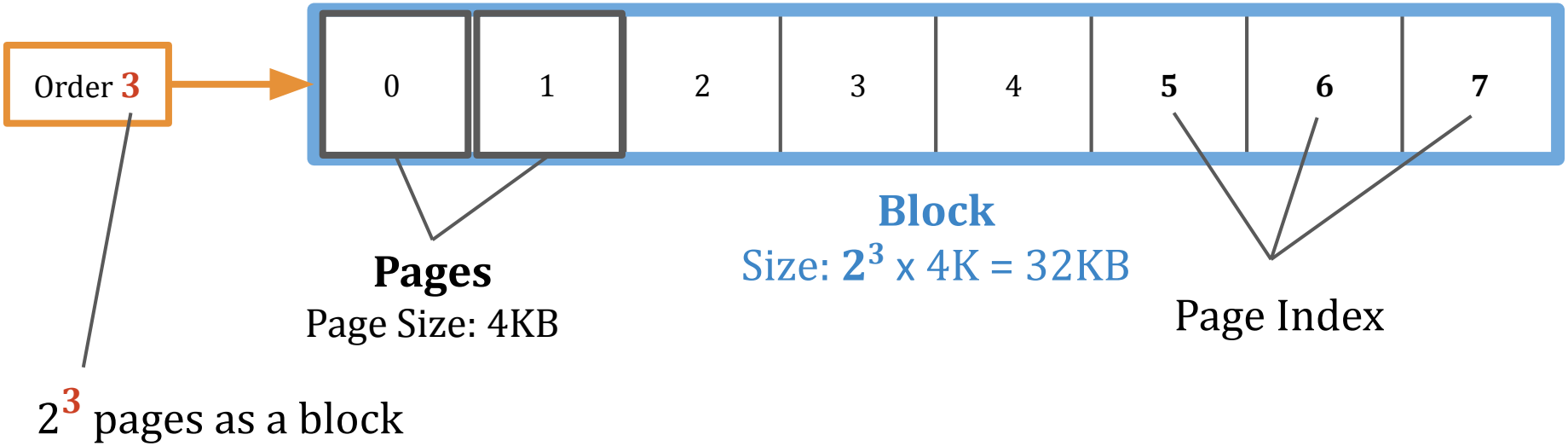
Buddy System

Buddy system is a well-known and simple algorithm for allocating contiguous memory blocks.

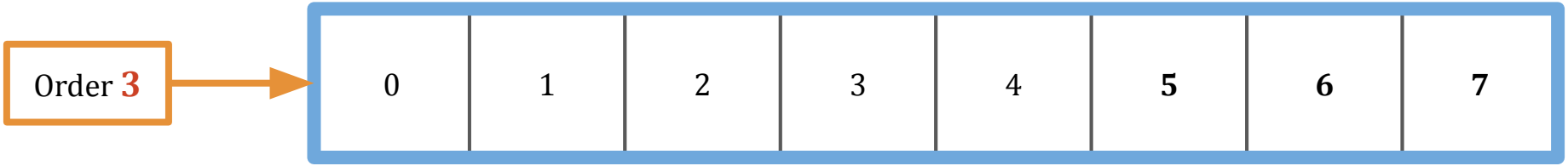


Buddy System

Buddy system is a well-known and simple algorithm for allocating contiguous memory blocks.

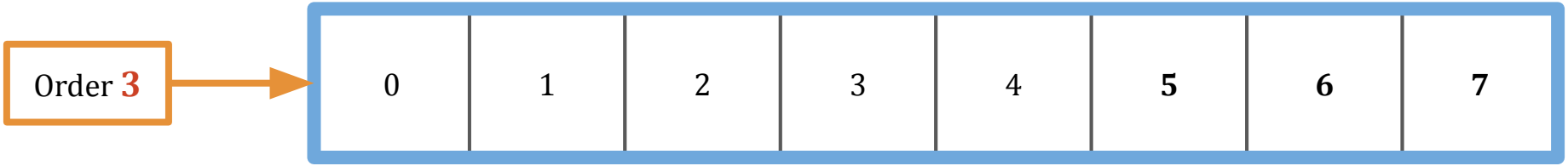


Buddy System — Example




```
void example() {
    p1 = allocate(2048);
    p2 = allocate(6144);
    free(p1);
    free(p2);
}
```

Buddy System — Example Allocate 2KB




```
void example() {
    p1 = allocate(2048);
    p2 = allocate(6144);
    free(p1);
    free(p2);
}
```

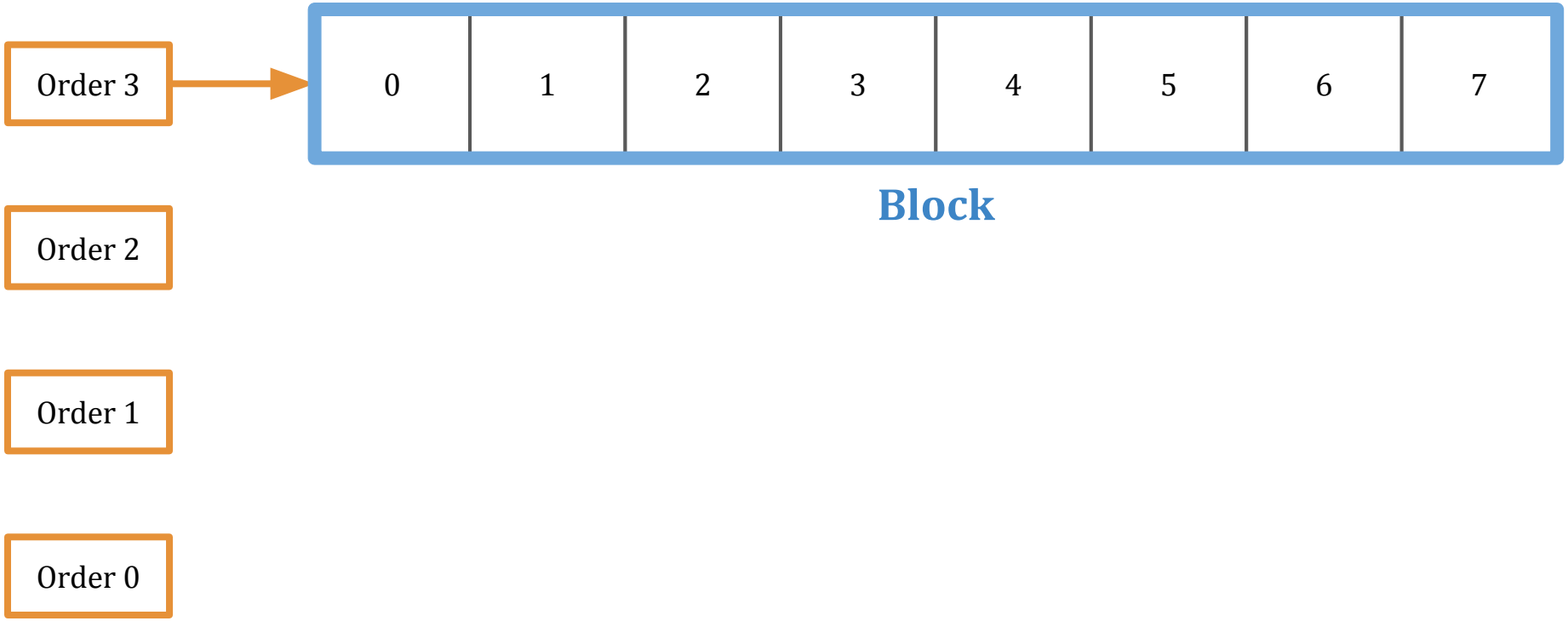
To allocate 2KB:

2KB $\leq$ 4KB (page size)

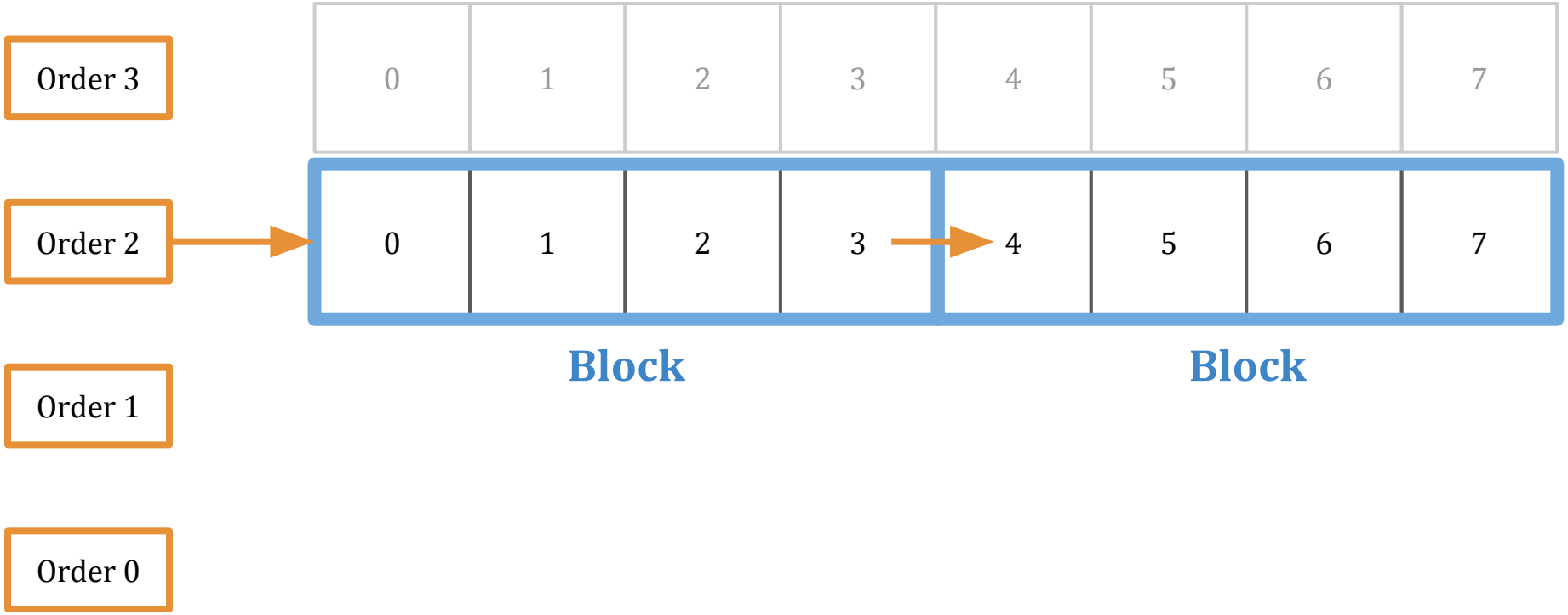
$\Rightarrow$ **1 page = 2^0 page**

$\Rightarrow$ Allocate a block at order **0**

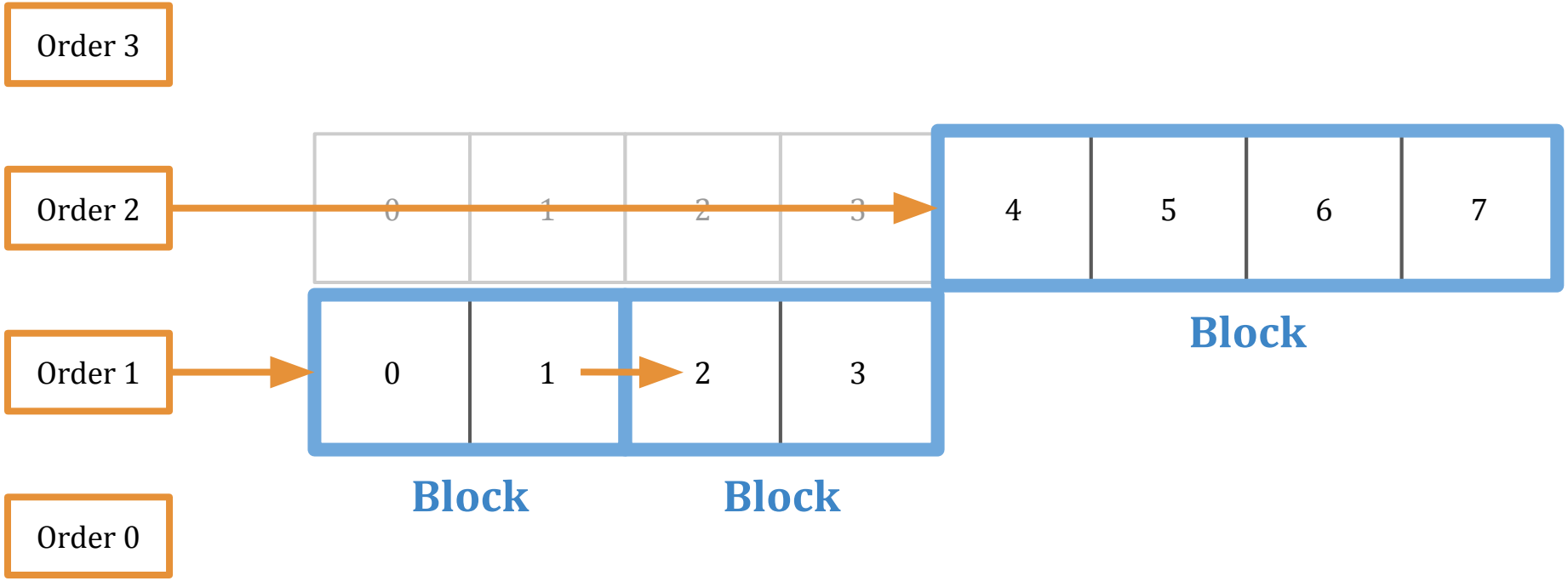
Buddy System — Example Allocate 2KB



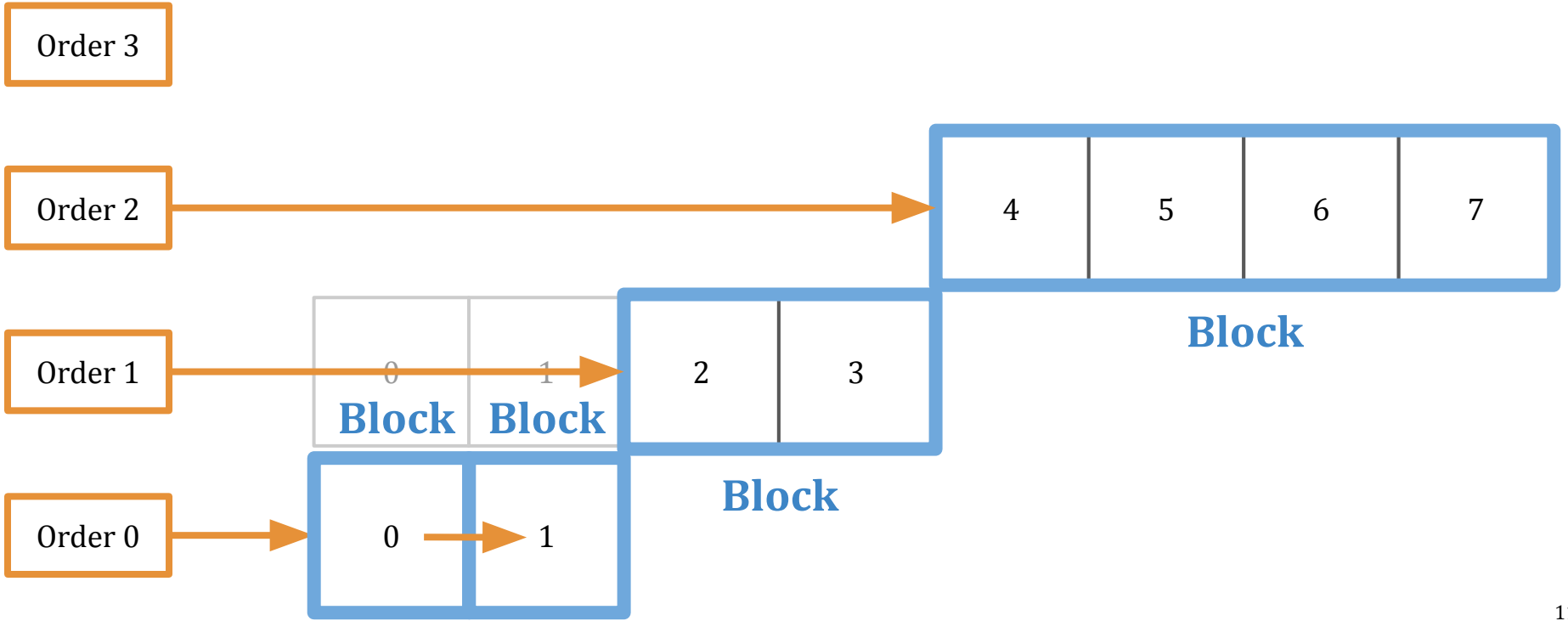
Buddy System — Example Allocate 2KB



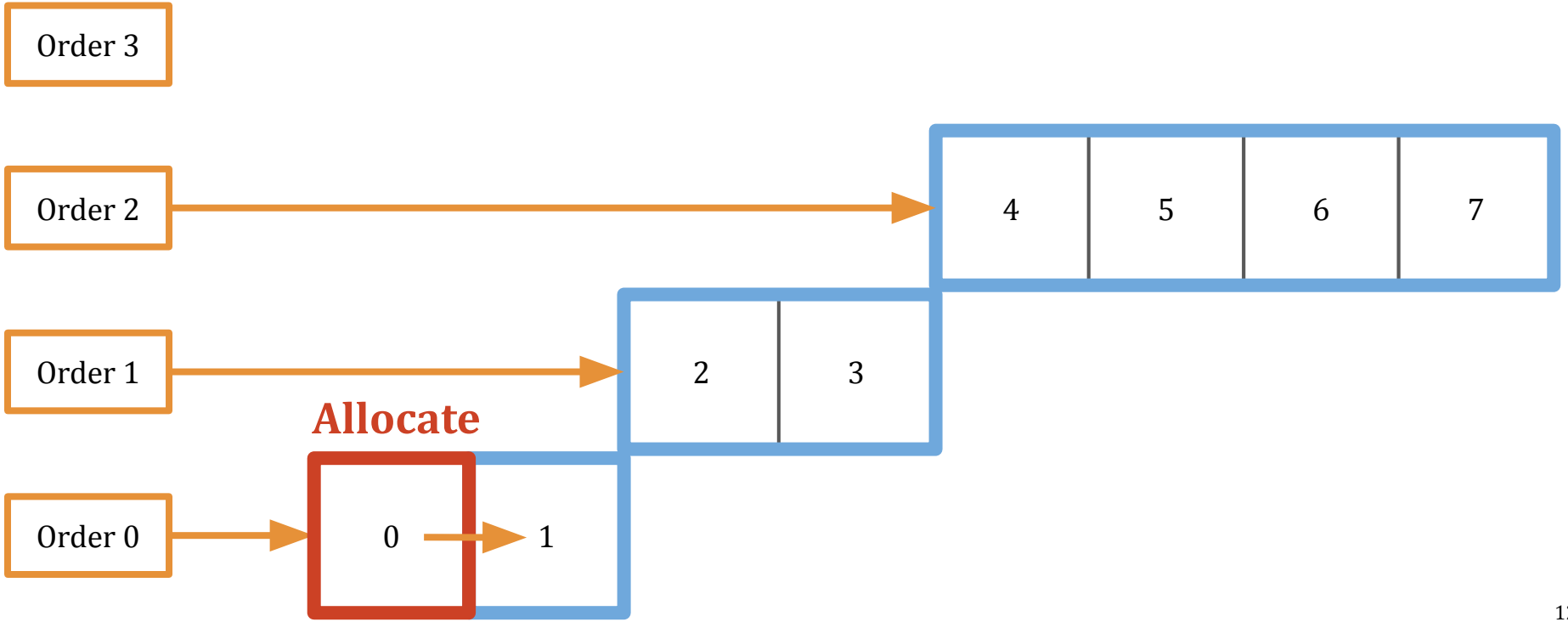
Buddy System — Example Allocate 2KB



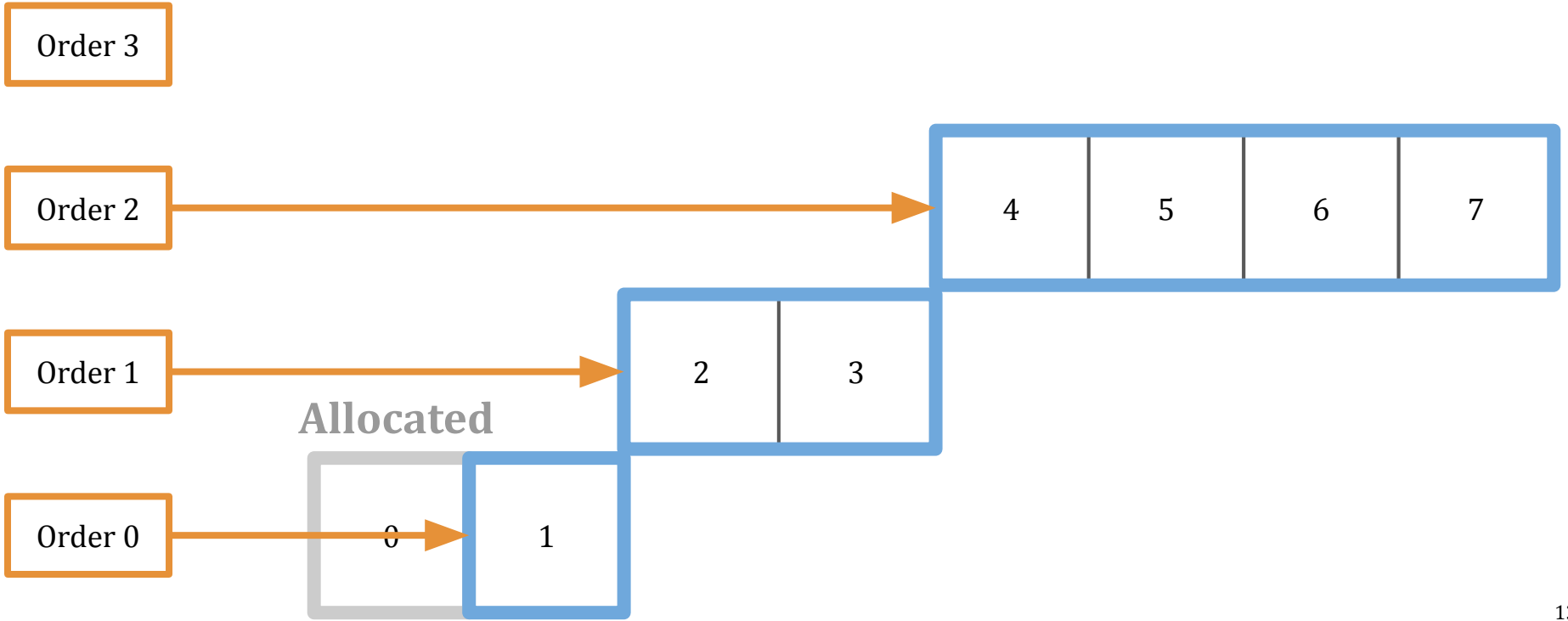
Buddy System — Example Allocate 2KB



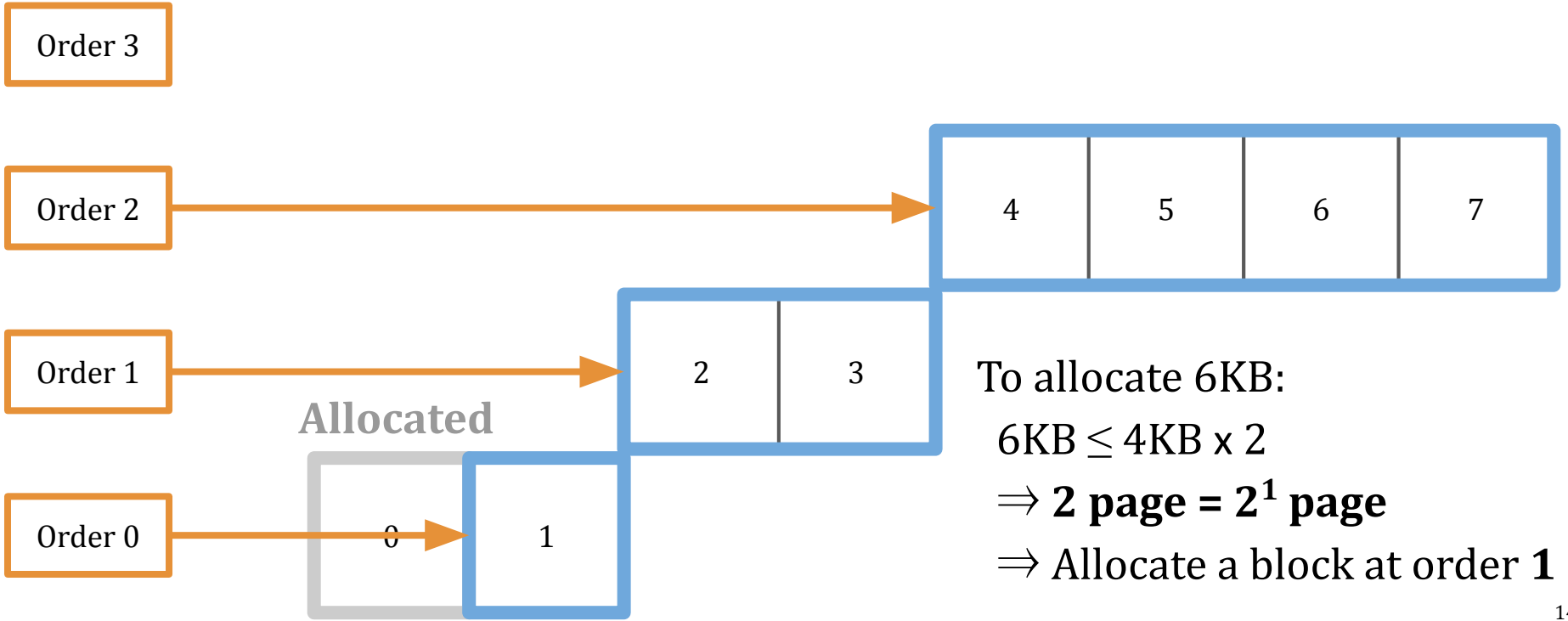
Buddy System — Example Allocate 2KB



Buddy System — Example Allocate 2KB



Buddy System — Example Allocate 6KB



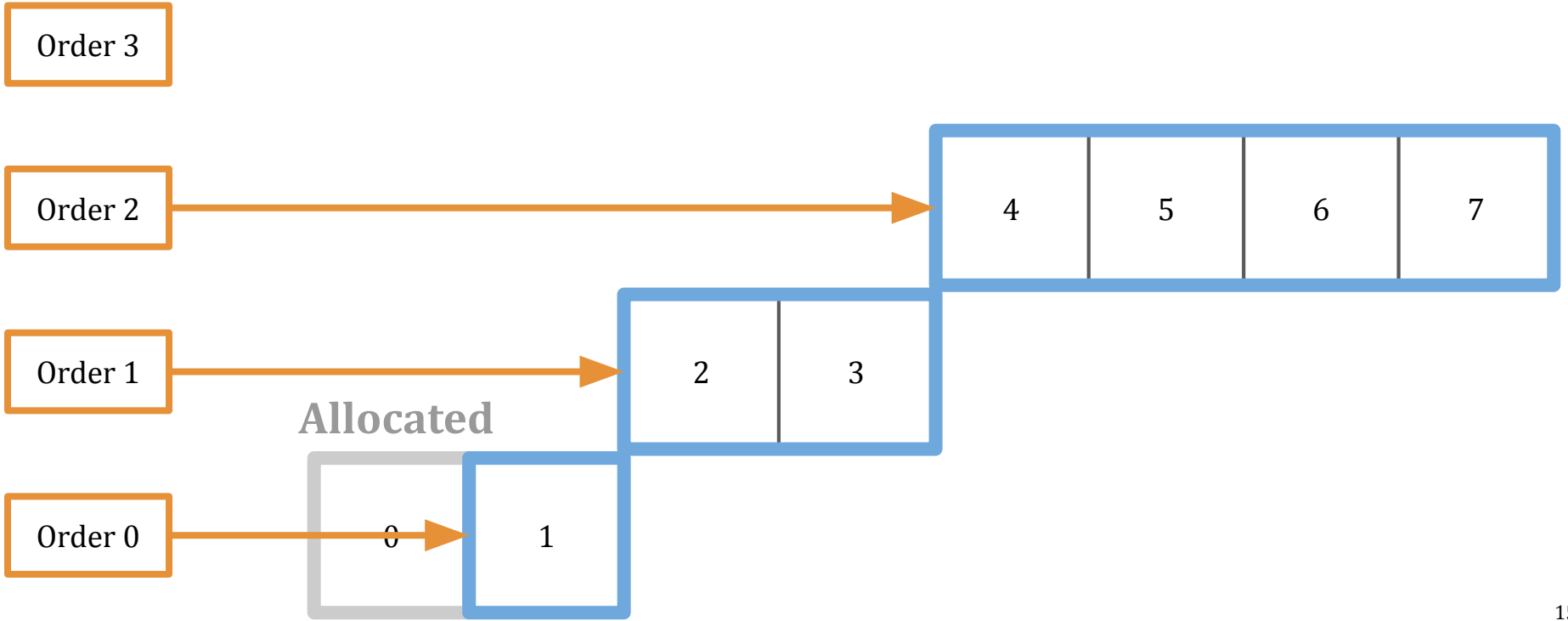
To allocate 6KB:

$6KB \leq 4KB \times 2$

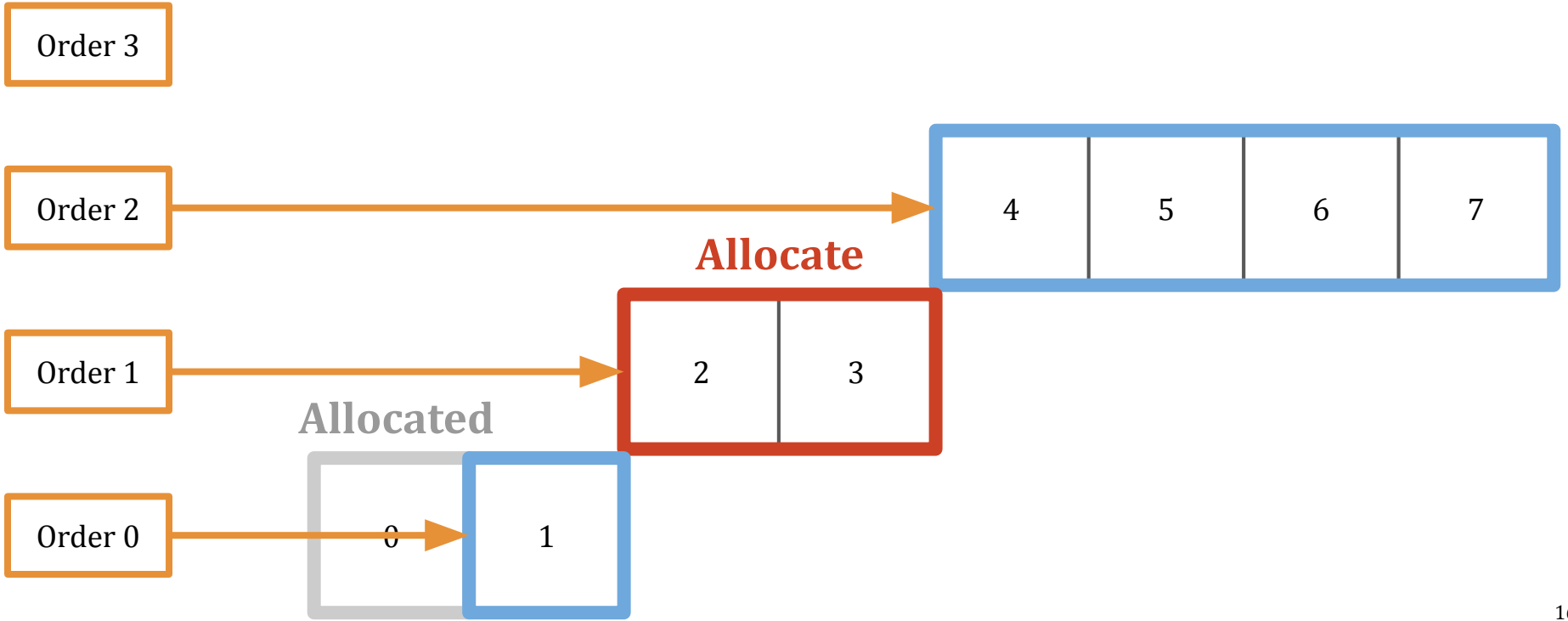
$\Rightarrow$ **2 page = 2^1 page**

$\Rightarrow$ Allocate a block at order **1**

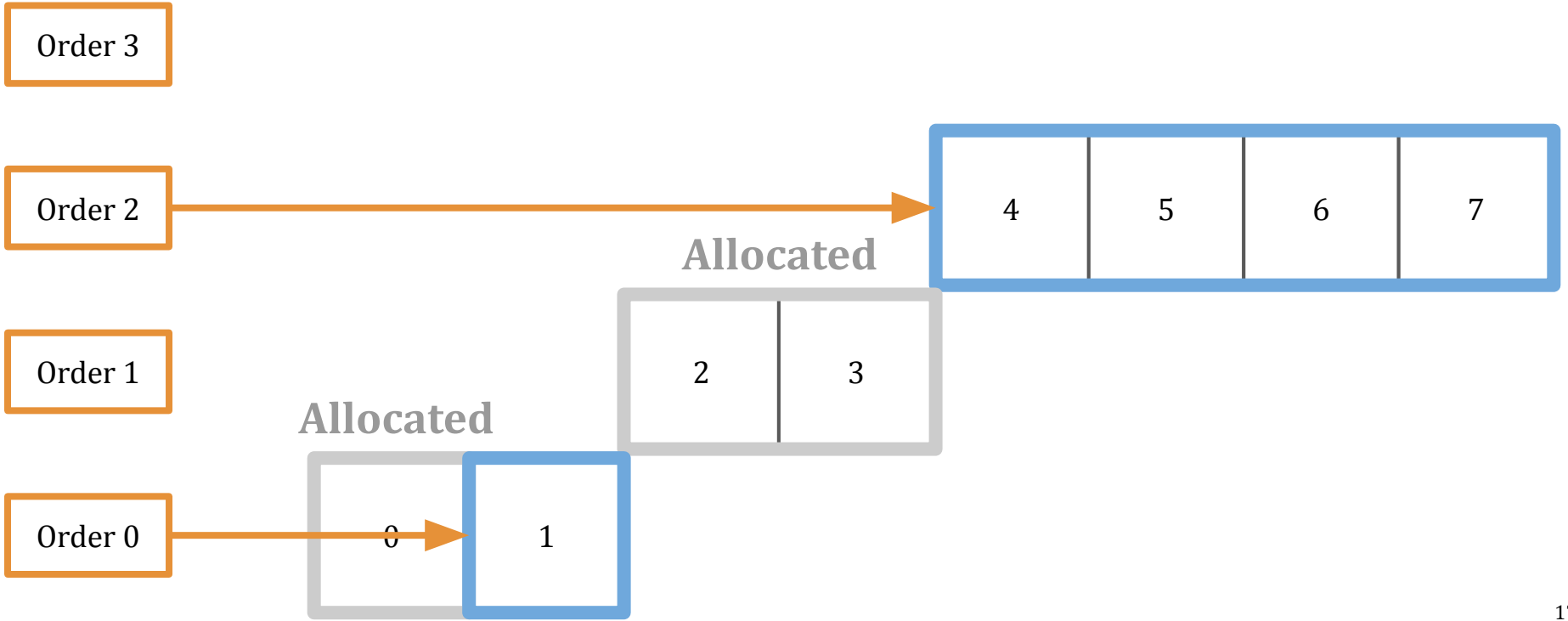
Buddy System — Example Allocate 6KB



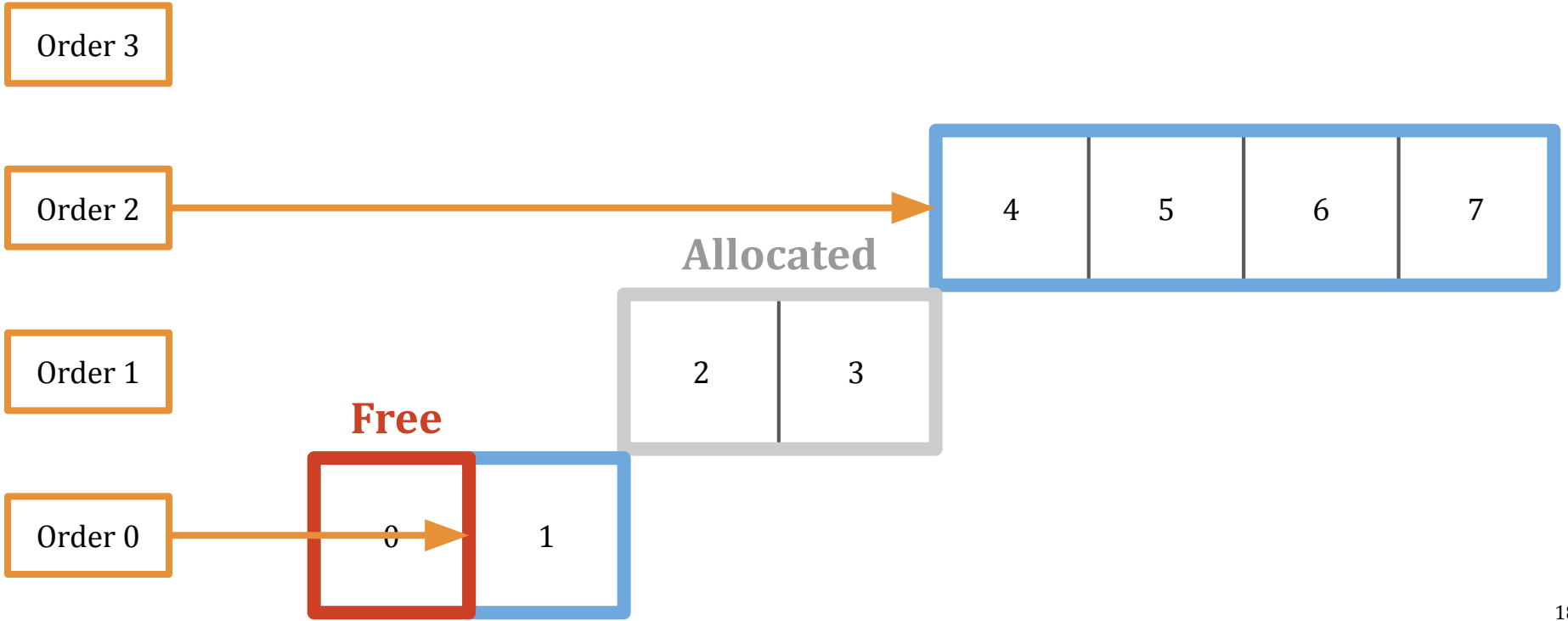
Buddy System — Example Allocate 6KB



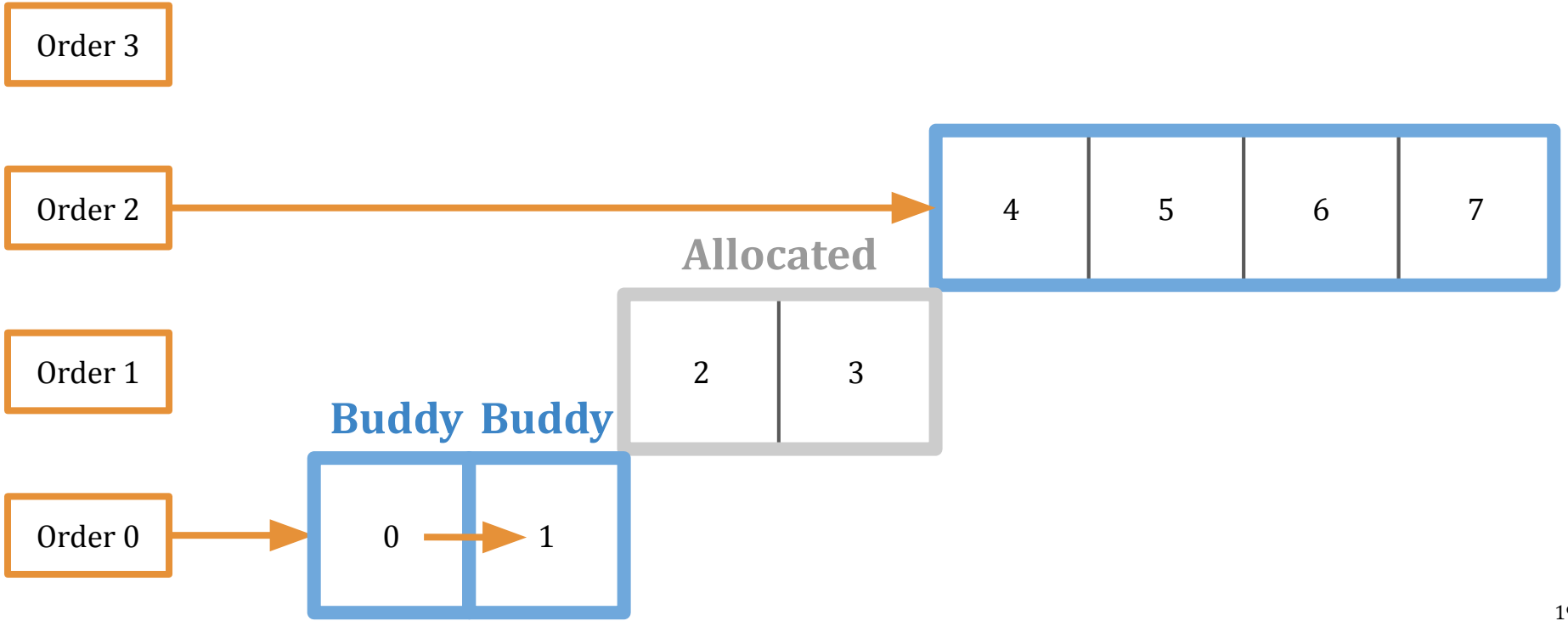
Buddy System — Example Allocate 6KB



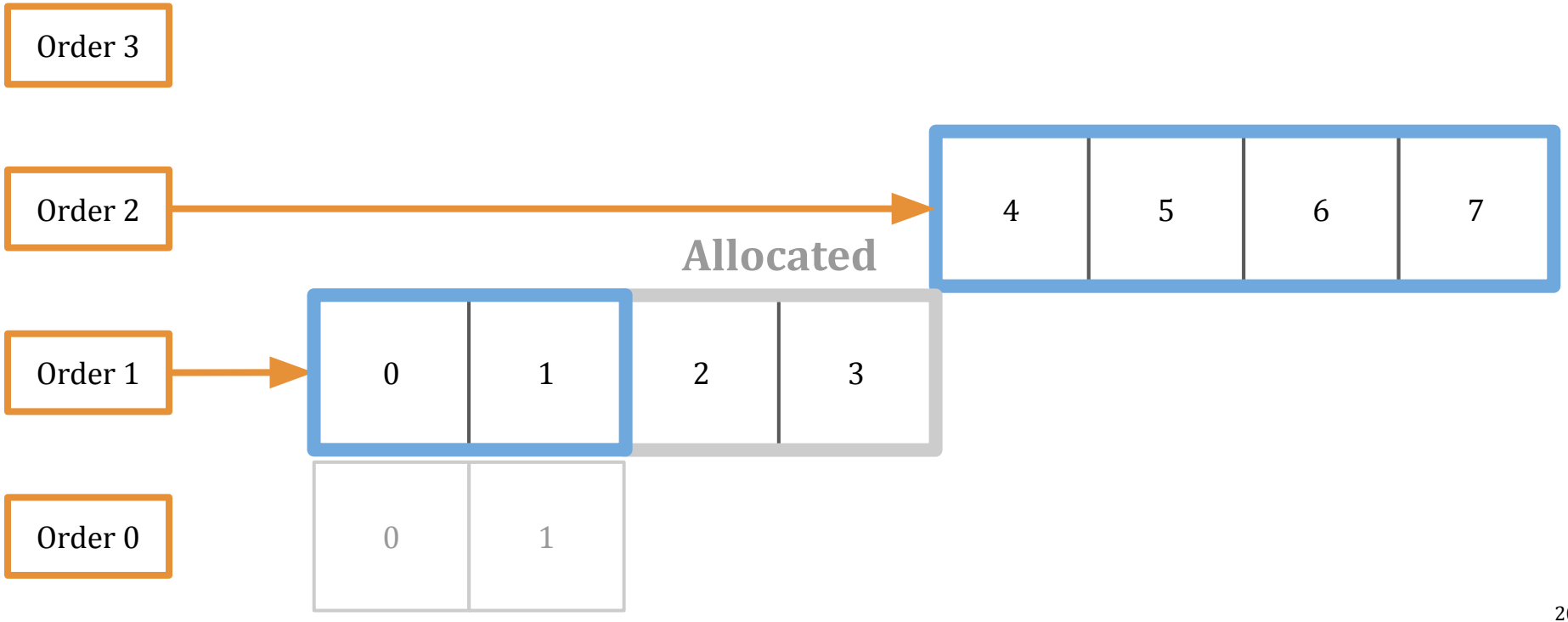
Buddy System — Example Free 2KB



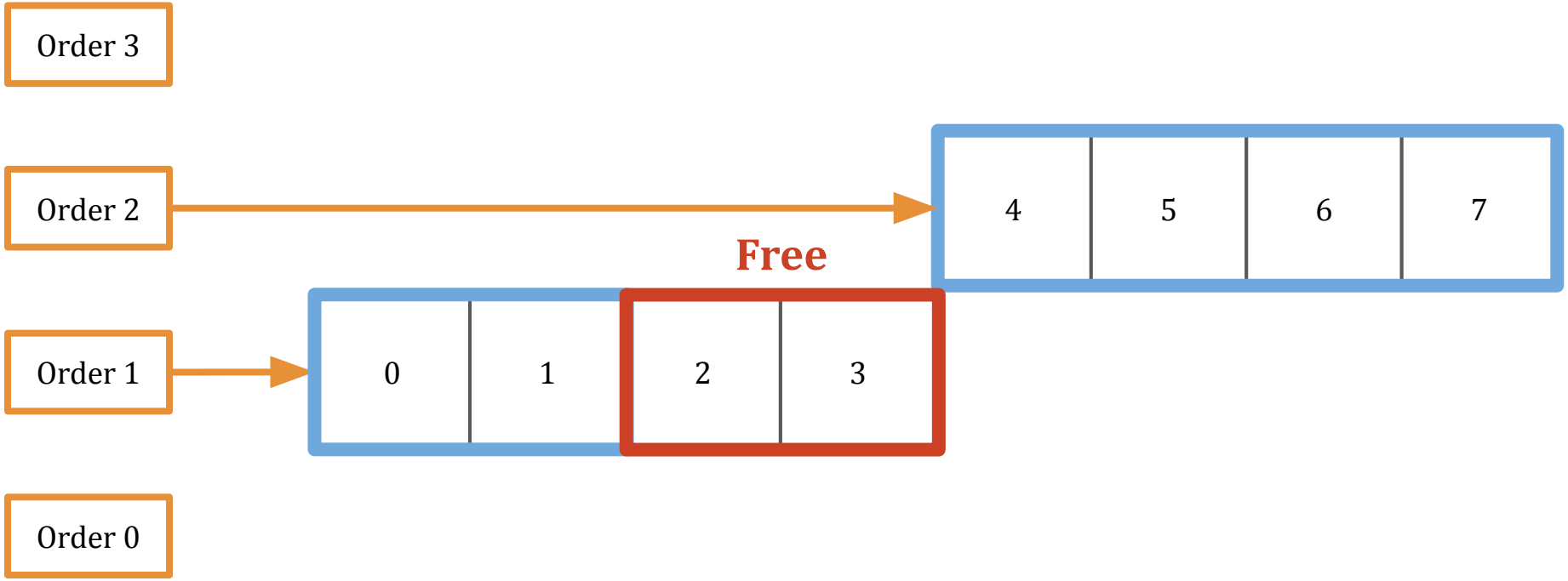
Buddy System — Example Free 2KB



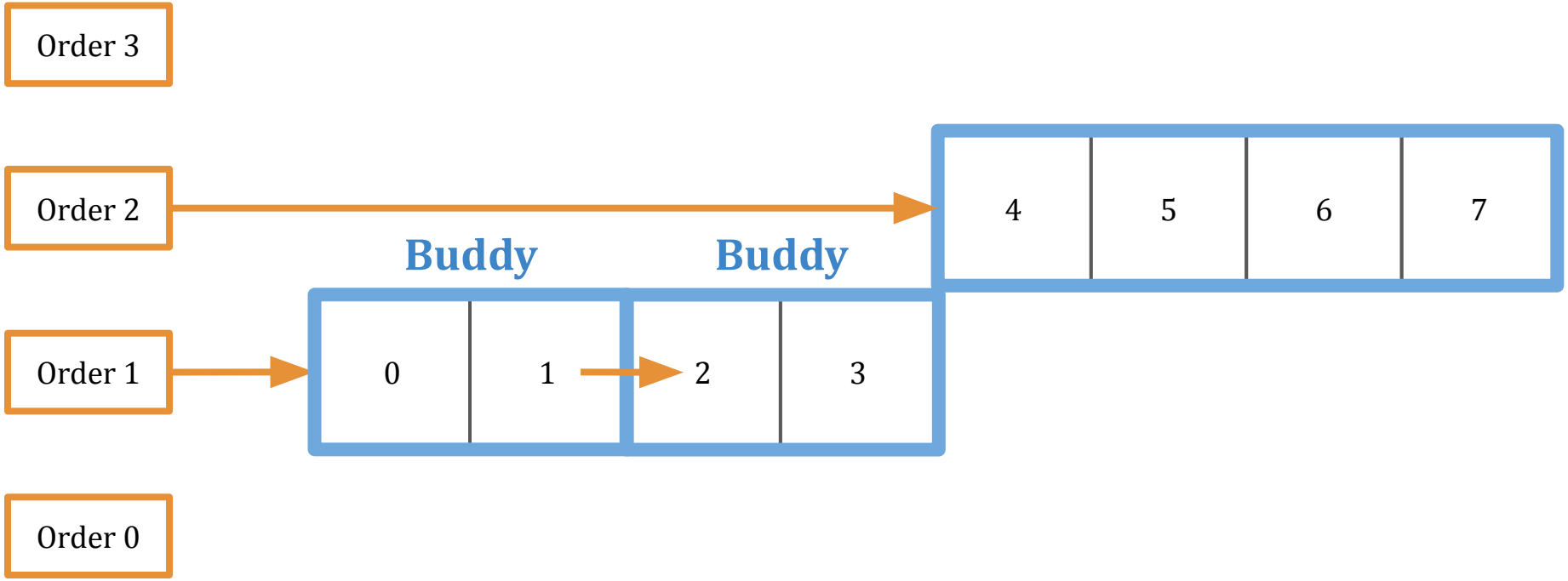
Buddy System — Example Free 2KB



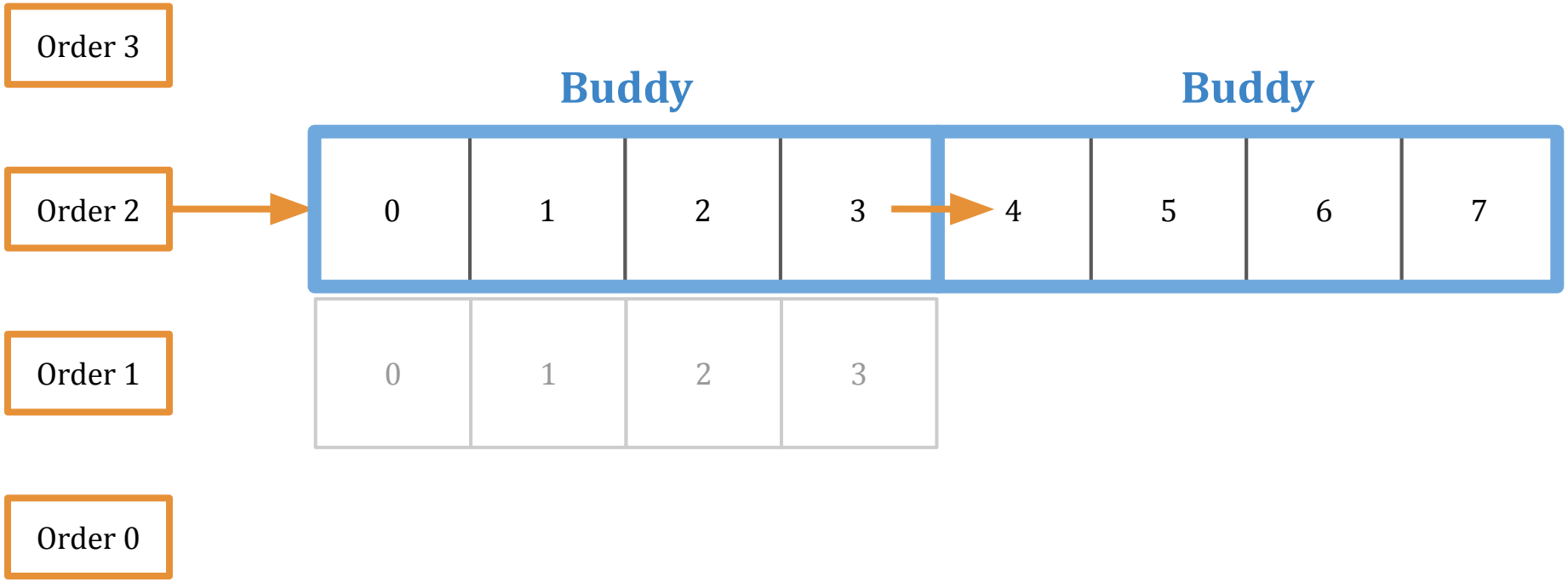
Buddy System — Example Free 6KB



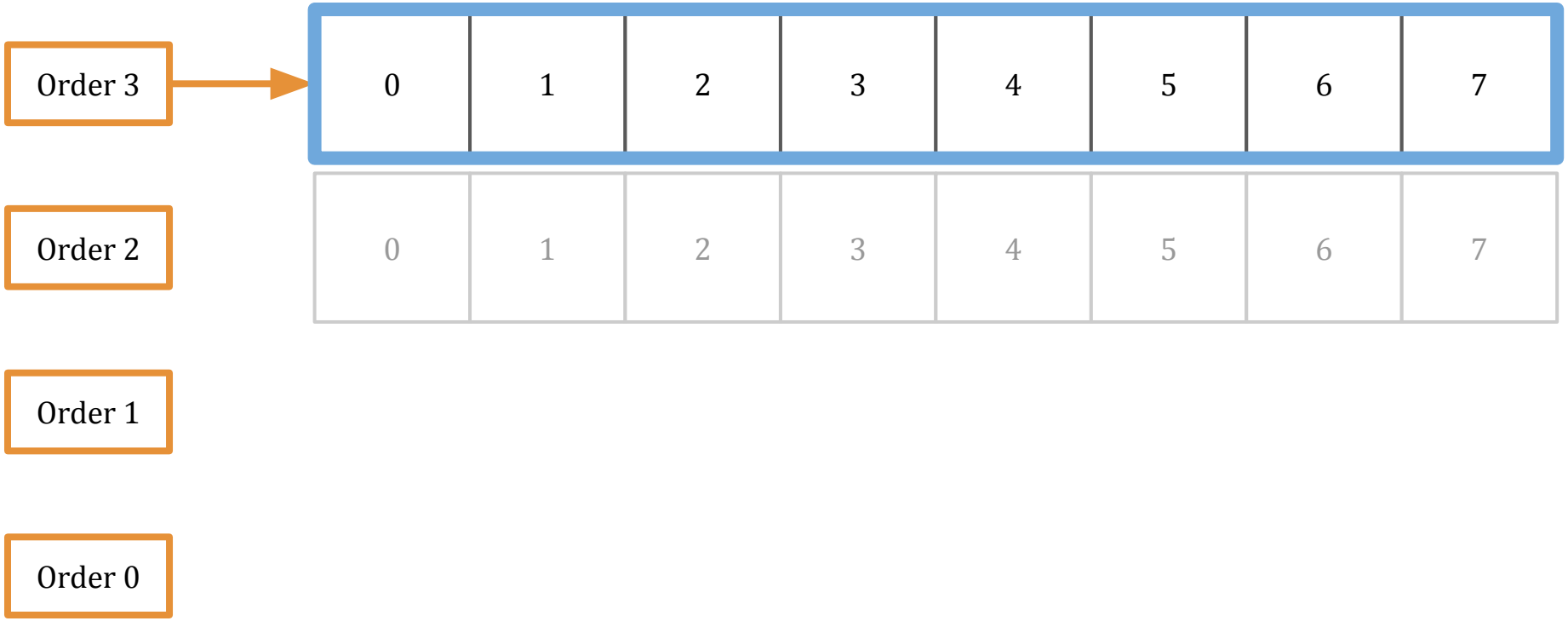
Buddy System — Example Free 6KB



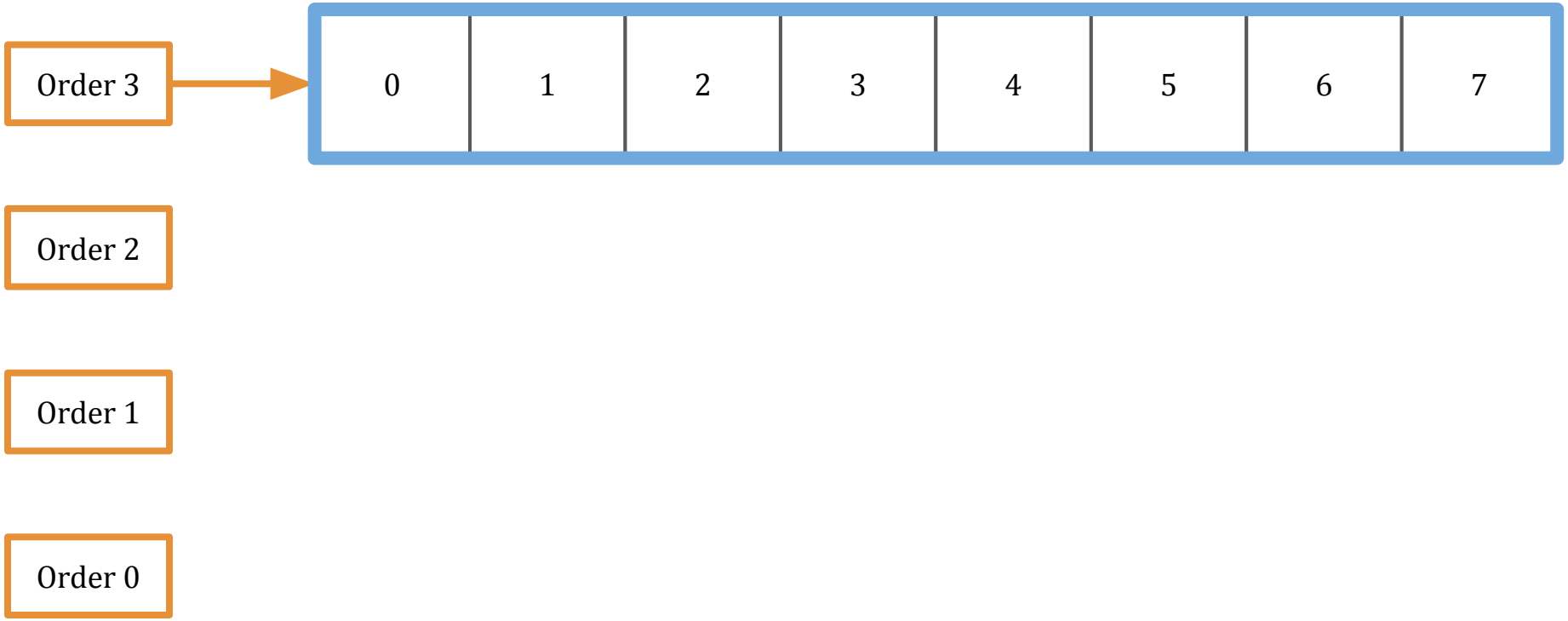
Buddy System — Example Free 6KB



Buddy System — Example Free 6KB



Buddy System — Example Free 6KB



Exercise 3-1:Implementation Logic

- **alloc_pages :**
  - Detach the leading block from the identified “free_area” list
  - Execute iterative bisection (split) if the located block's order exceeds the requested order
  - Append the residual split halves into the “free_area” lists
- **free_pages :**
  - Revert the target block's metadata to the unallocated state
  - Merge the target block and its buddy into a unified 2^{order+1} block

```
struct page* alloc_pages(unsigned int order) {
    // TODO: Implement this function
}

void free_pages(struct page* page) {
    // TODO: Implement this function
}
```

Exercise 3-1: Expected Result

```
(base) hsien@hsien-System-Product-Name:~/OSC/lab3/ex1$ make run
./prog

p1:
free_area[10] 2559
free_area[9] 1
free_area[8] 1
free_area[7] 1
free_area[6] 1
free_area[5] 1
free_area[4] 1
free_area[3] 1
free_area[2] 1
free_area[1] 1
free_area[0] 0

p2:
free_area[10] 2559
free_area[9] 1
free_area[8] 1
free_area[7] 1
free_area[6] 1
free_area[5] 1
free_area[4] 1
free_area[3] 1
free_area[2] 1
free_area[1] 0
free_area[0] 0
```

```
p3:
free_area[10] 2559
free_area[9] 1
free_area[8] 1
free_area[7] 1
free_area[6] 1
free_area[5] 1
free_area[4] 1
free_area[3] 1
free_area[2] 0
free_area[1] 1
free_area[0] 0

free:
free_area[10] 2560
free_area[9] 0
free_area[8] 0
free_area[7] 0
free_area[6] 0
free_area[5] 0
free_area[4] 0
free_area[3] 0
free_area[2] 0
free_area[1] 0
free_area[0] 0
```

Exercise 3-1: Answer

```
struct page* alloc_pages(unsigned int order) {
    if (order > MAX_ORDER) return nullptr;

    for (unsigned int current_order = order; current_order <= MAX_ORDER; ++current_order) {
        if (!free_area[current_order].empty()) {
            struct page* page = free_area[current_order].front();
            free_area[current_order].pop_front();

            while (current_order > order) {
                --current_order;
                size_t pfn = page - mem_map.data();
                size_t buddy_pfn = pfn + (1 << current_order);
                struct page* buddy = &mem_map[buddy_pfn];

                buddy->order = current_order;
                buddy->refcount = 0;
                free_area[current_order].push_back(buddy);
            }

            page->order = order;
            page->refcount = 1;
            return page;
        }
    }
    return nullptr;
}
```

Exercise 3-1: Answer

```
void free_pages(struct page* page) {
    if (!page || page->refcount == 0) return;

    size_t pfn = page - mem_map.data();
    unsigned int order = page->order;
    page->refcount = 0;

    while (order < MAX_ORDER) {
        size_t buddy_pfn = pfn ^ (1 << order);
        if (buddy_pfn >= NUM_PAGES) break;

        struct page* buddy = &mem_map[buddy_pfn];

        if (buddy->refcount != 0 || buddy->order != order) {
            break;
        }

        free_area[order].remove(buddy);

        if (buddy_pfn < pfn) {
            page = buddy;
            pfn = buddy_pfn;
        }

        order++;
        page->order = order;
    }

    free_area[order].push_back(page);
}
```

Exercise 3-2: Memory Reservation

- Mark specific physical memory regions (e.g., Kernel Image, DTB) as occupied to prevent the Buddy System from allocating them
- Top-Down Splitting
  - **No Overlap** : Block is safe and keep it in the free list
  - **Full Overlap** : Block falls entirely within the reserved region (Remove from the free list and set “**refcount** ” as allocated)
  - **Partial Overlap** : Split it into two equal child blocks, and push them to the next lower level (order - 1) for further evaluation

Exercise 3-2: Reserved Memory Regions

- The following memory regions must be reserved before any allocation:
  - Device Tree Blob (DTB)
  - Kernel image
  - Initramfs (ramdisk)
  - Any additional platform-specific reserved areas

```
void memory_reserve(phys_addr_t base, size_t size) {
    // TODO: Implement this function
}
```

Exercise 3-2: Expected Result

```
(base) hsien@hsien-System-Product-Name:~/OSC/lab3/ex2$ make run
rm -f prog
g++ -g -o prog main.cpp
./prog
free_area[10] 2037
free_area[9] 0
free_area[8] 1
free_area[7] 1
free_area[6] 0
free_area[5] 0
free_area[4] 1
free_area[3] 0
free_area[2] 1
free_area[1] 1
free_area[0] 0
```

TODOs

Exercise 3-1

- **alloc_pages :** Allocate a contiguous block of pages using the split logic
- **free_pages :** Release the specified page block and iteratively merge it with adjacent unallocated buddies

Exercise 3-2

- **memory_reserve :** Isolate specific physical memory regions from the allocator by marking their pages as allocated